STRUCTURE OF ATOM

Representation of atom: ${}_{z}X^{A}$

- A = mass Number
- z = atomic Number
- X = symbol of atom

Mass Number (A): Number of proton + Number of Neutron
- Known as nucleons
- always a whole NO.

Atomic Number (z):
- Neutrons = mass number − Atomic Number
- Known as nuclear charges
- Number of protons in nucleus

Different atomic species:

- Isotopes : same atomic NO. but different mass Number :- ${}^{B}_{A}X \quad {}^{C}_{A}Y \quad {}^{D}_{A}Z$
- Isobars : same mass NO. but different atomic NO. :- ${}^{B}_{A}X \quad {}^{C}_{A}Y \quad {}^{D}_{A}Z$
- Isodiaphers : constant or same (n-P) : ${}_{7}N^{15}$ & ${}_{9}F^{19}$
- Isotones : same Number of neutrones : ${}_{19}K^{39}$ & ${}_{20}Ca^{40}$
- Isosters : same Number of atom & electrons :- CO_2 & N_2O
- Isoelectronic :- same Number of electrones : BF_3 & SO_2

Wave Nature of electromagnetic radiation:-

- Electromagnetic wave do not require medium and can move in Vaccum
- In vaccum all EMW travel with speed of Light
- $c = \nu\lambda$, $\nu = \frac{1}{\lambda}$
  - c = speed of Light 3×10^8 (m/s)
  - ν = Frequency (Hz)
  - λ = Wave length (m)

Particle Nature of Electromagnetic radiatation: Plank's Quantum theory

- Energy is emitted or absorbed in form of quantum (Packets of energy)
- smallest energy packet of light is Known as Photon
- Energy of each quantum ∝ frequency of radiation

$E = \frac{hc}{\lambda}$, $E = \frac{nhc}{\lambda}$

- h = 6.67×10^{-34} J s
- n = No. of photons

Bohr's atomic model

- it is Based on quantum theory
- Applicable only For single e^- species
- Radius of orbit = $0.529 \frac{n^2}{Z}$ Å
- Energy of an e^- = $-21.69 \times 10^{-19} \frac{Z^2}{n^2}$ J/atom

  = $-13.6 \frac{Z^2}{n^2}$ eV/atom

- P.E = -2 K.E = $2E_T$

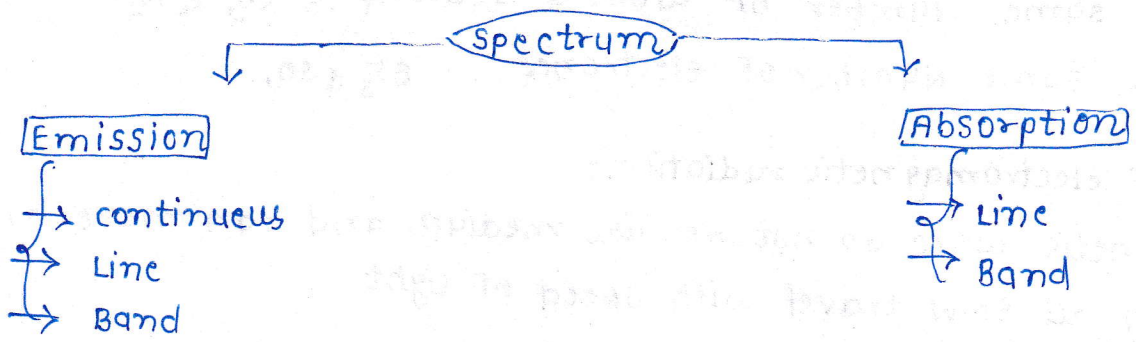
  KE = $-E_T$

Energy difference between two energy Levels: $-13.6\ Z^2 \left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$ eV/atom

velocity of an electron: $2.18 \times 10^6 \frac{Z}{n}$ m/s

SPECTRUM

when radiation passed through the prism then due to dispersion, Pattern obtain is called spectrum.



Hydrogen spectrum:

Series	n_1	n_2	No. of lines	Region
Lyman	1	2, 3, 4 ... ∞	$n_2 - 1$	UV
Balmer	2	3 4 5 .... ∞	$n_2 - 2$	visible
Paschen	3	4 5 6 ..... ∞	$n_2 - 3$	Infrared
BRacket	4	5 6 7 ..... ∞	$n_2 - 4$	Infrared
P. fund	5	6 7 8 ..... ∞	$n_2 - 5$	Infrared
Humphery	6	7 8 9 ..... ∞	$n_2 - 6$	Far Infrared

Rydberg Formula : $\bar{\nu} = \frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

- R = Rydberg constant = $109678\ cm^{-1}$
- $\frac{1}{R} = 9.12 \times 10^{-6}\ cm = 912\ \text{Å}$

De-Broglie concept : $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m(KE)}}$

Heisenberg's Uncertainty principle :- $\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$

- Δx = Uncertainity in positi
- Δv = Uncertainity in velocity
- ΔP = Uncertainity in momentu

Quantum Number

Possible location of an electron in an atom

1. principal Q.No.
2. Azimuthal Q.No.
3. Magnetic Q.No.
4. Spin Q.No.

① Principal Quantum Number :- Determine size & energy of orbital

- For n^{th} shell
  - No. of orbitals = n^2
  - No. of Subshell = n
  - No. of max. e^- = $2n^2$
  - Angular momentum = $nh/2\pi$
- $n \propto$ distance From nucleus ; ✱ : $r_1 < r_2 < r_3 < r_4$
  - $\propto$ Energy of shell : $E_1 < E_2 < E_3 < E_4$
  - $\propto \frac{1}{\text{velocity}}$: $v_1 > v_2 > v_3 > v_4$

② Azimuthal Quantum No :- Determine 3-D shapes of orbital

value of l = 0 to $n-1$

$n = 1 \leftarrow l = 0$

$n = 2 \leftarrow l = 0, 1$

$n = 3 \leftarrow l = 0, 1, 2$

value of l	subshell	shape
0	s	spherical
1	p	dumb bell shape
2	d	double dumbbell
3	f	complex

- For the value of l
  - No. of orbital $= 2l+1$
  - max e^- in subshell $= 2(2l+1)$
  - orbital angular momentum $= \frac{h}{2\pi}\sqrt{l(l+1)}$

③ Magnetic Quantum No :- Determine the orientation of orbital

value of m = $-l$ to $+l$ include zero

l	m
0	0
1	-1 0 +1
2	-2 -1 0 +1 +2

④ Spin Quantum No :- Determine the spin of the electron

value of $s = \pm \frac{1}{2}$

spin angular momentum of $e^- = \sqrt{s(s+1)} \cdot \frac{h}{2\pi}$

RULES For Filling of orbitals

1. Aufbau principle
2. Paulis principle
3. Hund's rule

1. Aufbau Principle :- orbitals are filled in their increasing energies

Lowest $(n+l)$ value orbital filled first

if $(n+l)$ is same then lowest value of n filled first

2. Pauli's Principle : No two e^- can have same set of four Q.No.

orbital accomodate max 2 e^- with opposite spin

3. Hunds Rule :- paining start when only every subshell get at least one electron with same spin